

Wazuh SIEM Home Lab — Attack Simulation & Detection

A self-built, segmented home lab used to deploy and administer a Wazuh SIEM for real-time log monitoring, incident triage, and troubleshooting across a multi-OS network — built to practice the day-to-day work of a SOC Analyst.



Architecture Diagram

Overview

- **Environment:** VirtualBox, 3-node isolated network
- **SIEM:** Wazuh Manager centralizing log collection from all agents
- **Hosts:** Ubuntu/Debian, CentOS/RHEL agents + one isolated "monitored" host with no internet access
- **Goal:** Simulate a small monitored enterprise environment and practice detection, triage, and troubleshooting end-to-end — not just install-and-forget.

What I Built

1. **Deployed and administered a Wazuh SIEM manager** to centralize log collection and monitor the 3-node network for security events.
2. **Configured remote syslog ingestion (UDP/514)** — including editing `ossec.conf` and

troubleshooting XML configuration errors — to enable real-time log forwarding from monitored hosts.

3. Built and segmented a multi-OS lab network

(Debian/Ubuntu, CentOS/RHEL) with static IP addressing to simulate a realistic monitored environment.

4. Applied defense-in-depth principles by isolating the vulnerable/monitored host from

internet access, using controlled internal file transfer for any tooling that needed to reach it.

Problems I Ran Into (and Fixed)

Log-forwarding pipeline failure

Logs stopped reaching the manager from one agent. I used `tcpdump` for packet-level traffic analysis, checked running processes on the agent, and audited firewall/iptables rules to isolate where the pipeline was breaking — traced it back to a blocked outbound rule.

Agent-manager compatibility issues

Ran into architecture mismatches and cross-version protocol conflicts between agents and the manager. Resolved this through manual, key-based agent enrollment instead of the default auto-enrollment flow.

Skills Practiced

Wazuh SIEM administration syslog tcpdump iptables
Linux (Ubuntu, CentOS/RHEL)

network segmentation incident triage root-cause
troubleshooting

Notes

This repo documents a personal lab project used for learning,
not a production deployment.
Configuration snippets are sanitized/simplified for sharing.